Ascites

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ASCITES



Definition

Ascites is defined as the accumulation of free fluid in the peritoneal cavity.

Causes

- Most often caused by liver cirrhosis (>75%)

- The remaining 25 % is due to

malignancy (10%),

heart failure (3%),

TB (2%),

pancreatitis (1%),

other rare causes.

Prognosis

- Most common complication of cirrhosis

- An indication of deterioration in clinical status and poor prognosis (e.g. refractory ascites and SBP)

- Two-year survival of patients with ascites is approximately 50%

- Cirrhotic patients develop ascites need liver transplantation in 10 years duration.

Pathophysiology

1- Portal hypertension (renal sodium & water retention)

2- Hypoalbuminaemia

3- Increased hepatic and splanchnic lymph production and transudation

1- Portal hypertension

↓

Vasodilatation, Decrease Splanchnic Systemic vascular resistance

↓

Reduction in arterial blood volume

↓

Activation of neurohumoral pressor systems

↓

Renal sodium & water retention

Diagnosis

I) Clinical:

The International Ascites Club grading (2003)

Grade 1: mild ascites “detectable only by US”.

Grade 2: moderate ascites manifested by moderate symmetrical abdominal distension.

Grade 3: large ascites with marked abdominal distension.

II) Paracentesis

- Gross Appearance of Ascitic Fluid

Color	Appearance
Translucent or yellow	Normal/sterile
Brown	Hyperbilirubinemia
Cloudy or turbid	Infection
bloody	Malignancy, Abdominal trauma

- **Ascitic Fluid Testing:** Cell count & differential, Albumin, Culture, Cytology, SAAG

Treatment Options

- Diet

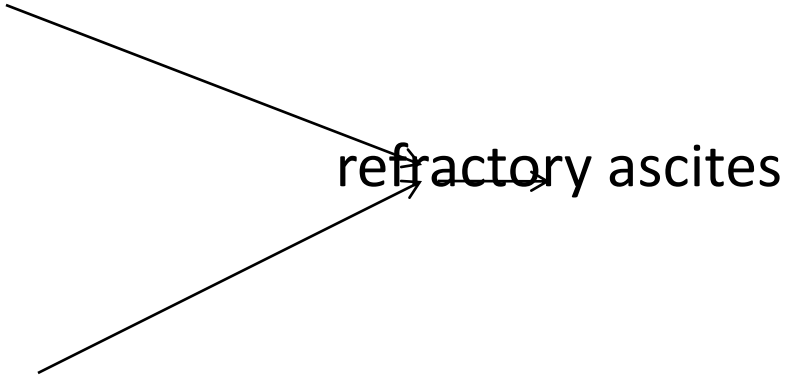
- Dietary sodium restriction
  - Limiting sodium intake 80-120 mmol/l (4.6–6.9 g) /day

Diuretics

- The most successful therapeutic regimen is the combination of

  <u>Spironolactone</u> and <u>Furosemide</u>

- Paracentesis
- Salt free albumin
- TIPS
- Shunts
- Liver transplant



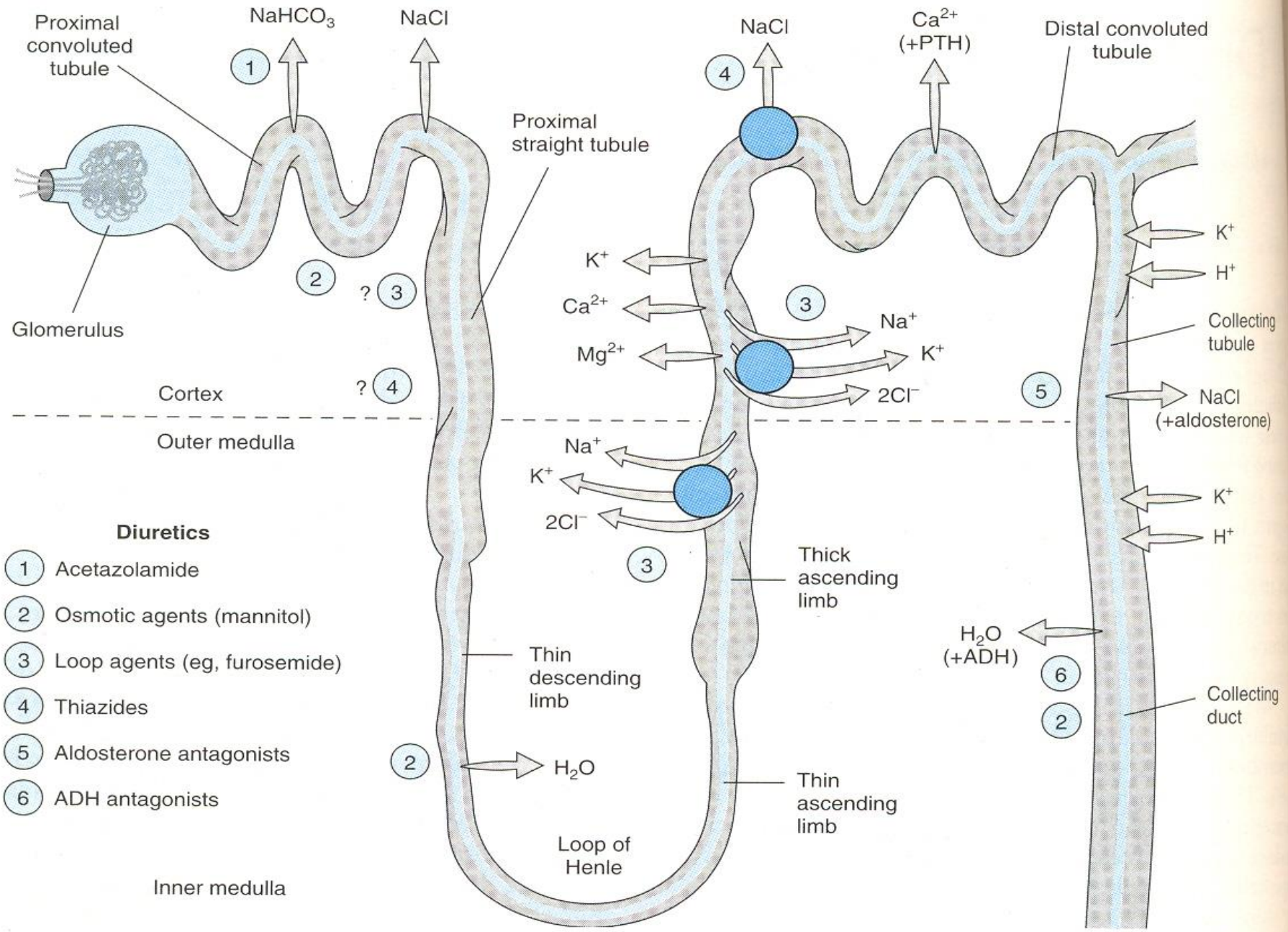


Figure 15–1. Tubule transport systems and sites of action of diuretics.

Loop Diuretics

- Furosemide, Bumetanide, Torsemide

Potassium Sparing Diuretics

- **<u>Aldosterone receptor antagonist</u>** **(**Antimineralocorticoids**)**

  Spironolactone,

- **<u>Non-aldosterone antagonist</u>**

  Amiloride, Triamterene

Spironolactone

- Spironolactone is drug of choice
- Aldosterone antagonist acting in distal tubule to increase natriuresis and conserve potassium
- Initial dose 100 mg and increasing up to 400 mg every 3-5 days up to 7 days (~ 3 days)
- Better natriuresis and diuresis than a loop diuretic
- Gynaecomastia, renal impairment, hyperkalaemia frequently limits the use
- Change to Amiloride if painful or disfiguring gynecomastia develops

Frusemide

- Use either

  - only if 400 mg spironolactone fails to achieve weight loss

or - single morning oral doses of Spironolactone and Furosemide, beginning with 100 mg and 40 mg (up to 400 mg Spironolactone & 160 mg Frusemide)

- Start at 40 mg a day and increasing 40 mg every 3rd day to max of 160 mg

- S.E : hyponatraemia, hypokalaemia, hypovolemia, renal impairment

Assess response to diuretics:

- Weight loss of 0.5 kg/day in absence of oedema and 1 kg/day when oedema present

  - Watch out for electrolyte disturbance

  - Over-diuresis is associated with intravascular volume depletion → renal impairment, hepatic encephalopathy and hyponatraemia

- Medical treatment based on sodium restricted diet, diuretics – response in 90 % without renal failure.

Spontaneous Bacterial Peritonitis

- Infection of ascitic fluid without an evident intra-abdominal surgically treatable source (in patients with liver cirrhosis)
- Clinical manifestations: Fever, abdominal pain, abdominal tenderness, confusion
- The diagnosis is established by
  - A positive ascitic fluid bacterial culture (mono-microbial)
  - Elevated ascitic fluid absolute polymorphonuclear leukocyte (PMN) count (>250 cells/mm^3)

SBP – Antibiotic Therapy

- Initiate for PMN ≥ 250/mm^3
- IV Cefotaxime 2g/8 hrs or Ceftriaxone 2g/24hrs
- Other drugs e.g.

  = Ticarcillin-clavulanate 3.1 g IV q6h **or**

  = Piperacillin-tazobactam 3.375 g IV q6h or 4.5 g IV q8h **or**

  = Ampicillin-sulbactam 3 g IV q6h **or**

  = Ertapenem 1 g IV q24h **or**

  = Levofloxacin 500 mg IV q24h **or**

  = Moxifloxacin 400 mg IV q24h

- Duration of therapy
  - 2 weeks suggested if Blood cultures (+)
  - If repeat paracentesis at 48 hrs shows PMN ≤ 250/mm^3, then 5-7 days of treatment may be adequate

- **<u>IV albumin (on day 1 and 3)</u>** for patients with

  - serum creatinine >1 mg/dl, or

  - BUN >30 mg/dl

  - total bilirubin >4 mg/dl.

  (1.5 g/kg within 6 hrs of enrollment and repeated 1.0 g/kg on day 3 as an adjunctive to cefotaxime).

- NB: Probiotic therapy in conjunction with antimicrobial treatment????

SBP – prophylaxis (Prophylactic antibiotics)

- Prescribed indefinitely until ascites has eliminated
- Prevent SBP in high-risk groups:

  - Patients with ascites admitted with acute GI bleeding

  - Patients with ascitic fluid protein levels of less than 1 g/dl

  - Patients with a prior episode of SBP

- Options include

  - Norfloxacin: 400 mg daily - Ciprofloxacin: 750 mg weekly

  - Five doses of double-strength trimethoprim-sulfamethoxazole per week (Monday through Friday)

NB: long-term norfloxacin subsequently developed quinolone-resistant spontaneous bacterial peritonitis

- **For patients with acute GIT bleeding**,

IV ceftriaxone 1 g IV daily is recommended for a total duration of 7 days (to decrease risk of infections, re-bleeding, and mortality). Or

once patients are stable with control of bleeding and resumption of oral intake, they may be transitioned to oral norfloxacin 400 mg twice daily to complete the 7-day course.

THANK YOU